

Ankush Jain

Backend Engineer

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SUMMARY

Backend engineer with five years on distributed systems for regulated healthcare and imaging - EHR integration, cloud infrastructure, access control and AI pipelines. Owns systems end to end: design, implementation, security hardening and the CI/CD that ships them.

EXPERIENCE

Apra Labs

Software Engineer

May 2023 – Present

Engineering services; client work across healthcare, medical devices and imaging. Clients not named.

Clinical Trial Management Platform (Nov 2024 – Present)

- Built the integration hub's base adapter modules, turning each new EHR integration from a bespoke build into a repeatable pattern; implemented the first adapter as the reference implementation
- Designed and shipped a full role-based access control system across two applications, covering admin and basic access - admin-only endpoint restriction, per-role response filtering, and pre-authorization checks
- Re-architected inter-service connectivity on managed pub/sub with async subscriptions, dead-letter handling and retry enhancement
- Cut the backend deployment pipeline from 26 min 33 s to 18 min 07 s (-32%), shortening every developer's feedback loop

Regenerative Medicine Data Platform (May 2026 – Present)

- Modelled clinical protocols and surveys on FHIR resources with OMOP/LOINC terminology, and implemented change logs purely through FHIR provenance
- Drove requirements with the client directly - converting whiteboard sessions and slide feedback into wireframed workflows and user stories

Document AI / Handwriting Recognition Pipeline (May 2025 – Nov 2025)

- Reduced the deployable image from 19 GB to 5.28 GB (-72%) through deliberate packaging design
- Designed the codebase so the third-party model packages and deploys with the system and its licence can be swapped on expiry, instead of requiring a rebuild
- Produced the high- and low-level design independently with minimal supervision

Headless Video Recording Platform (Apr 2024 – Oct 2024)

- Wrote the Node backend from scratch and connected it to Redis, the C++ core and the frontend, giving the system one consistent source of truth
- Took the recorder from an unstable state to a dependable one, debugging intermittent segmentation faults in frame decoding across edge hardware

AR Surgical Navigation (Aug 2023 – Apr 2024)

- Identified the signed distance field computation as the pipeline's bottleneck and devised a compute-once-and-reuse technique, saving over 80% of the time it took per frame
- Onboarded two junior engineers and led the project through a stretch when senior engineers were unavailable, having become its most familiar contributor

SKILLS

Languages: Python, C++, Java, Kotlin, Go, JavaScript/Node.js, TypeScript

Backend & messaging: Spring Boot, Ktor, REST APIs, Server-Sent Events (SSE), RabbitMQ, Google Cloud Pub/Sub, Event-driven architecture, Microservices

Databases: PostgreSQL, SQL, Redis, Flyway (schema migrations), Database design, Cloudflare D1

Cloud & DevOps: Google Cloud Platform (GCP), Cloud Run, Terraform, Docker, CI/CD pipelines, Infrastructure as code (IaC), AWS (exposure)

Healthcare & standards: HL7 FHIR, OMOP, LOINC, SMART on FHIR, EHR integration, Clinical trial systems

Security: Role-based access control (RBAC), Authentication & authorization, OAuth / token-based auth, Microsoft Graph API, Static analysis remediation

ML & imaging: Computer vision, Image registration, Signal processing (FFT/DFT), Hyperspectral imaging, OCR / handwriting recognition, VTK

EDUCATION

Dr. SPM IIIT Naya Raipur — B.Tech, Computer Science and Engineering

2017 – 2021